Joseph Lee

Computational Biologist | Machine Learning for Biological Data | RNA-seq, Protein Modelling, Python
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SUMMARY

Computational biologist specialising in machine learning applications to biological data, with experience across RNA-seq modelling, protein-ligand prediction models and large-scale genomic datasets. Experience in building and evaluating models using Python-based frameworks, including analysis of 20,000+ genes and performance benchmarking under realistic biological constraints. Combines industrial experience in therapeutic discovery with computational expertise to translate biological problems into predictive insights for drug discovery.

SKILLS

Machine Learning & Modelling: Scikit-learn, TensorFlow, Bayesian modelling, Gaussian Processes
Programming: Python (NumPy, pandas, matplotlib, seaborn), R, SQL
Structural Bioinformatics: AlphaFold, Chai, Boltz, PyMol, MolProbity
Data management & Infrastructure: HPC, Docker, Unix, Git

EXPERIENCE

Postgraduate Researcher 05/2025 – 09/2025
Lovell Lab, University of Manchester

- Evaluated ML-based protein-ligand models under low-homology conditions, identifying performance limitations relevant to real-world drug discovery
- Benchmarked Chai and Boltz across 18 protein-ligand systems, identifying a consistent false-positive pattern, where chemically and sterically implausible ligands were predicted as viable binders, undermining screening reliability and motivating incorporation of ligand-aware constraints to improve model specificity

Postgraduate Researcher 01/2025 – 05/2025
Rattray Group, University of Manchester

- Optimised a Bayesian Machine Learning approach for time-series analysis of expression patterns for ~20,000 genes (scRNA-seq), identifying ~4,450 previously undetected expression patterns from traditional models
- Visualised and translated statistical outputs and assumptions for wet-lab scientists to inform downstream gene validation experiments

R&D Analyst (part-time) 06/2025 – PRESENT
Patent Outsourcing Ltd

- Delivered patent and literature analyses across UK, EU, and US jurisdictions
- Translated complex data into actionable insights for biotech clients, explaining complex concepts to non-specialists

Associate Scientist (Protein Engineering) 01/2021 – 08/2023
Immunocore

- Engineered and screened 1200+ antibody candidates to optimise target specificity across immuno-oncology programmes
- Statistical analysis and evaluation of internal drug databases (3000+ molecules) to benchmark high-potency molecules and support candidate selection for oncology
- Identified cross-departmental automation improvements, saving ~6 hours per week of manual colony picking to accelerate experimental throughput

PERSONAL PROJECTS

QSAR Lite: Molecular Solubility Prediction | Python, scikit-learn, DeepChem, Docker, Git, hyperparameter tuning

- Developed a machine learning model using gradient boosted trees to predict small molecule solubility, enabling reproducible deployment using Docker

Proteomic modelling pipeline | Python, BASH

- Built a proteomic modelling pipeline to process bacterial genomic sequences end-end (ORF prediction, peptide digestion, mass/ion analysis, visualisation), enabling scalable analysis and interpretability of downstream proteomics outputs

EDUCATION

University of Manchester 09/2024 – 09/2025
MSc Bioinformatics & Systems Biology **Grade: Distinction (79%)**

- Modules: Programming (84%), Computational Biology (81%), Research Project 2 (80%), Research Project 1 (78%), Bioinformatics (68%), Statistics (Pass/Fail) (Pass)
- **Activities**: German Society

University of Bath 09/2017 – 06/2021
BSc (Hons) Molecular and Cellular Biology **Grade: 2:1 (67%)**

- Dissertation (71%): Statistical analysis of 120+ publicly available SARS-CoV-2 datasets to assess population-wide intranasal vaccination strategies
- **Activities**: Italian (CEFR level A2), University of Bath Cricket Club: Secretary, University of Bath Rugby Club